Unit 2

Lesson Three

De Moivre's theorem

De Moivre's theorem

For any complex number $z = r(\cos\theta + i\sin\theta)$ where θ is the principle amplitude of z

1 De Moivre's theorem with positive integer power

If n is an integer power then $z^n = r^n(\cos n\theta + i\sin n\theta)$

For example :

If $z = 3(\cos 30° + i\sin 30°)$

Then $z^4 = 3^4\left(\cos(4\times 30°) + i\sin(4\times 30°)\right)$

$= 81(\cos 120° + i\sin 120°)$

2 De Moivre's theorem with positive rational power

If k is a positive integer , θ is measured in sixtieth measure then :

$z^{\frac{1}{k}} = r^{\frac{1}{k}}\left(\cos\left(\frac{\theta + 360°\, n}{k}\right) + i\sin\left(\frac{\theta + 360°\, n}{k}\right)\right)$ where $n = 0, 1, 2, \ldots, (k-1)$ and if θ is measured in radian , use 2π instead of 360° and the resulting numbers by substitution $n = 0, 1, 2, \ldots (k-1)$ are called the kth root of the number z

For example :

If $z = 8(\cos 30° + i\sin 30°)$

then $z^{\frac{1}{3}} = 8^{\frac{1}{3}}(\cos 30° + i\sin 30°)^{\frac{1}{3}}$

$= 2\left(\cos\frac{30° + 360°\, n}{3} + i\sin\frac{30° + 360°\, n}{3}\right)$ where $n = 0, 1, 2$

at $n = 0 : z_1 = 2\left(\cos\frac{30°}{3} + i\sin\frac{30°}{3}\right) = 2(\cos 10° + i\sin 10°)$

at $n = 1 : z_2 = 2\left(\cos\frac{30° + 1\times 360°}{3} + i\sin\frac{30° + 1\times 360°}{3}\right) = 2(\cos 130° + i\sin 130°)$

at $n = 2 : z_3 = 2\left(\cos\frac{30° + 2\times 360°}{3} + i\sin\frac{30° + 2\times 360°}{3}\right) = 2(\cos 250° + i\sin 250°)$

$= 2\left(\cos(-110) + i\sin(-110)\right)$

i.e. $z^{\frac{1}{3}}$ has 3 values z_1 , z_2 and z_3 each is called the cubic root of the number z

Remark

The kth roots of the complex number can be obtained directly such that the amplitude of each of them $\left(\frac{\theta + 360\ n}{k}\right) \in]-\pi , \pi]$ putting $n = \cdots , -2 , -1 , 0 , 1 , 2 , \cdots$

Where if :

1. $\theta \in]0 , \pi]$

i.e. It is positive , put $n = 0 , \textcircled{-1} , 1 , -2 , 2 , -3 , \ldots$ to k values

"**Notice :** after zero we begin by the negative value"

2. $\theta \in]-\pi , 0]$

i.e. It is negative or zero , put $n = 0 , \textcircled{1} , -1 , 2 , -2 , 3 , \ldots$ to k values

"**Notice :** after zero we begin by the positive value"

Example 1

If $z = 2 + 2\sqrt{3}\ i$ find : 1. z^4 **2.** $z^{\frac{1}{4}}$

Solution

$\because z = 2 + 2\sqrt{3}\ i$ $\quad \therefore x = 2 \ , \ y = 2\sqrt{3}$

$\therefore |z| = r = \sqrt{x^2 + y^2} = \sqrt{4 + 12} = 4$

$\because x > 0 \ , \ y > 0$ $\quad \therefore z$ lies in the 1st quadrant

$\therefore \theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{2\sqrt{3}}{2}\right) = \tan^{-1}\left(\sqrt{3}\right) = 60°$

$\therefore z = 4\ (\cos 60° + i \sin 60°)$

1. $z^4 = 4^4\left(\cos (4 \times 60°) + i \sin (4 \times 60°)\right)$

$= 256\ (\cos 240° + i \sin 240°)$

$= 256\left(\cos (-120°) + i \sin (-120°)\right)$

2. $z^{\frac{1}{4}} = 4^{\frac{1}{4}}(\cos 60° + i \sin 60°)^{\frac{1}{4}}$

$= \sqrt{2}\left(\cos \frac{60° + 360° n}{4} + i \sin \frac{60° + 360° n}{4}\right)$ where $n = 0 , -1 , 1 , -2$

- at $n = 0$, then $z_1 = \sqrt{2}\ (\cos 15° + i \sin 15°)$
- at $n = -1$, then $z_2 = \sqrt{2}\left(\cos (-75°) + i \sin (-75°)\right)$
- at $n = 1$, then $z_3 = \sqrt{2}\ (\cos 105° + i \sin 105°)$
- at $n = -2$, then $z_4 = \sqrt{2}\left(\cos (-165°) + i \sin (-165°)\right)$

i.e. $z^{\frac{1}{4}}$ has 4 values z_1 , z_2 , z_3 and z_4 and each is called 4th root of z

Example 2

Find the trigonometric form of the expression $\left(\sqrt{3}+i\right)^{\frac{2}{5}}$

Solution

Let $z=\sqrt{3}+i$ $\quad\therefore x=\sqrt{3}\ ,\quad y=1$

$\therefore |z|=r=\sqrt{x^2+y^2}=\sqrt{3+1}=2$

, $\because x>0\ ,\quad y>0$ $\quad\therefore$ z lies in the 1st quadrant

$\therefore \theta=\tan^{-1}\left(\frac{y}{x}\right)=\tan^{-1}\left(\frac{1}{\sqrt{3}}\right)=30°$

$\therefore z=2(\cos 30°+i\sin 30°)$

$$, \because z^{\frac{2}{5}}=(z^2)^{\frac{1}{5}}=\left[\left(2(\cos 30°+i\sin 30°)\right)^2\right]^{\frac{1}{5}}$$

$$=[4(\cos 60°+i\sin 60°)]^{\frac{1}{5}}=4^{\frac{1}{5}}(\cos 60°+i\sin 60°)^{\frac{1}{5}}$$

$$=\sqrt[5]{4}\left(\cos\frac{60°+360°\,n}{5}+i\sin\frac{60°+360°\,n}{5}\right)\text{ where } n=0\ ,1\ ,2\ ,3\ ,4$$

at $n=0$: $\therefore z_1=\sqrt[5]{4}(\cos 12°+i\sin 12°)$

at $n=1$: $\therefore z_2=\sqrt[5]{4}(\cos 84°+i\sin 84°)$

at $n=2$: $\therefore z_3=\sqrt[5]{4}(\cos 156°+i\sin 156°)$

at $n=3$: $\therefore z_4=\sqrt[5]{4}(\cos 228°+i\sin 228°)=\sqrt[5]{4}(\cos(-132°)+i\sin(-132°)$

at $n=4$: $\therefore z_5=\sqrt[5]{4}(\cos 300°+i\sin 300°)=\sqrt[5]{4}\left(\cos(-60°)+i\sin(-60°)\right)$

$\therefore \left(\sqrt{3}+i\right)^{\frac{2}{5}}$ has five values $z_1\ ,z_2\ ,z_3\ ,z_4\ ,z_5$

The nth roots

The equation $x^n=a$, where a is a complex number has n roots in the form $x=a^{\frac{1}{n}}$ and these roots lie on one circle where they are represented by Argand's diagram of radius $|a^{\frac{1}{n}}|$ and form vertices of regular polygon of n sides.

Remember

$1=\cos 0+i\sin 0$

$i=\cos 90°+i\sin 90°$

$-1=\cos 180°+i\sin 180°$

$-i=\cos(-90°)+i\sin(-90°)$

Example 3

Represent on Argand's diagram the roots of the equation $z^5+32=0$

Solution

$\because z^5+32=0$ $\quad\therefore z^5=-32$ $\quad\therefore z=(-32)^{\frac{1}{5}}$

$\therefore z=(32\times -1)^{\frac{1}{5}}$ $\quad\therefore z=32^{\frac{1}{5}}(\cos 180°+i\sin 180°)^{\frac{1}{5}}$

$\therefore z = 2\left(\cos\frac{180° + 360° n}{5} + i \sin\frac{180° + 360° n}{5}\right)$

where $n = 0 , -1 , 1 , -2 , 2$

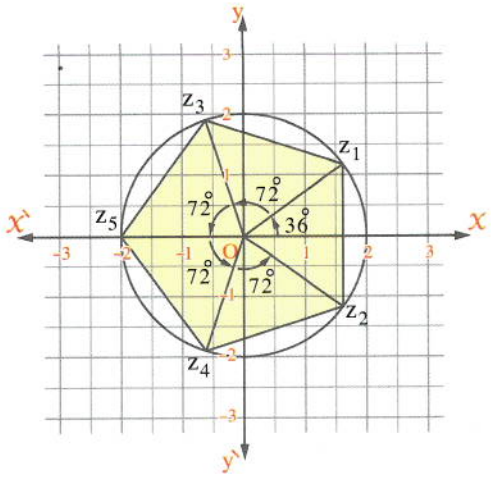
at $n = 0$: $\therefore z_1 = 2(\cos 36° + i \sin 36°)$

at $n = -1$: $\therefore z_2 = 2\left(\cos(-36°) + i \sin(-36°)\right)$

at $n = 1$: $\therefore z_3 = 2\left(\cos(108°) + i \sin(108°)\right)$

at $n = -2$: $\therefore z_4 = 2\left(\cos(-108°) + i \sin(-108°)\right)$

at $n = 2$: $\therefore z_5 = 2\left(\cos(-180°) + i \sin(-180°)\right)$



Remark

The roots of the equation are represented on Argand's diagram by vertices of regular pentagon lie on one circle whose centre is the origin (O) and its radius length is 2 length units such that $m(\angle z_n Oz_{n+1}) = \frac{360°}{5} = 72°$ and the 1st root z_1 can be calculated by substitution $n = 0$, so $z_1 = 2(\cos 36° + i \sin 36°)$ then add or subtract 72° to the amplitude of z_1 to get the amplitude of z_2 and so on to get the other roots.

Example 4

Find the two square roots of the complex number $z = 2 - 2\sqrt{3}\,i$ in algebraic form.

Solution

$\because z = 2 - 2\sqrt{3}\,i$ $\quad\therefore x = 2 \ , \ y = -2\sqrt{3}$

$\therefore |z| = r = \sqrt{x^2 + y^2} = \sqrt{4 + 12} = 4$

$, \because x > 0 \ , \ y < 0$ $\quad\therefore z$ lies in the 4th quadrant

$\therefore \theta = \tan^{-1}\left(\frac{y}{x}\right) = \tan^{-1}\left(\frac{-2\sqrt{3}}{2}\right) = \tan^{-1}\left(-\sqrt{3}\right) = -60°$

$\therefore z = r(\cos\theta + i \sin\theta) = 4\left(\cos(-60°) + i \sin(-60°)\right)$

$\therefore z^{\frac{1}{2}} = 4^{\frac{1}{2}}\left(\cos(-60°) + i \sin(-60°)\right)^{\frac{1}{2}}$

$= 2\left(\cos\left(\frac{-60° + 360° n}{2}\right) + i \sin\left(\frac{-60° + 360° n}{2}\right)\right)$ where $n = 0 , 1$

at $n = 0$: $\therefore z_1 = 2\left(\cos(-30°) + i \sin(-30°)\right) = 2\left(\frac{\sqrt{3}}{2} + i \times -\frac{1}{2}\right) = \sqrt{3} - i$

at $n = 1$: $\therefore z_2 = 2(\cos 150° + i \sin 150°) = 2\left(\frac{-\sqrt{3}}{2} + i \times \frac{1}{2}\right) = -\sqrt{3} + i$

$\therefore$ The two square roots of the number z are $\pm\left(\sqrt{3} - i\right)$

Another solution : (using the algebraic form without converting to the trigonometric form)

Assume $z^{\frac{1}{2}} = x + y\,i$ where $x , y \in \mathbb{R}$

$\therefore \left(2 - 2\sqrt{3}\,i\right)^{\frac{1}{2}} = x + y\,i$ by squaring both sides $2 - 2\sqrt{3}\,i = x^2 - y^2 + 2xy\,i$

$\therefore x^2 - y^2 = 2$ (1) , $2xy = -2\sqrt{3}$ and so $y = \frac{-\sqrt{3}}{x}$ (2)

by substitution from (2) in (1) :

$\therefore x^2 - \frac{3}{x^2} = 2$ (Multiply $\times x^2$) $\therefore x^4 - 3 = 2x^2$

$\therefore x^4 - 2x^2 - 3 = 0$ $\therefore (x^2 - 3)(x^2 + 1) = 0$

$\therefore x^2 + 1 = 0$ and so $x = \pm i$ $\left(\text{Refused because } \pm i \notin \mathbb{R}\right)$

or $x^2 - 3 = 0$ and so $x = \pm\sqrt{3}$ by substitution in (2)

$\therefore$ at $x = \sqrt{3}$ $\therefore y = \frac{-\sqrt{3}}{\sqrt{3}} = -1$ $\therefore z^{\frac{1}{2}} = \sqrt{3} - i$

at $x = -\sqrt{3}$ $\therefore y = \frac{-\sqrt{3}}{-\sqrt{3}} = 1$ $\therefore z^{\frac{1}{2}} = -\sqrt{3} + i$

$\therefore$ The two square roots of the number z are $\pm\left(\sqrt{3} - i\right)$

Example 5

If $x + y\,i = \left(\frac{1 + 4\sqrt{3}\,i}{1 - 4\sqrt{3}\,i}\right)^{\frac{1}{2}}$, find the real value of x , y

Solution

By squaring both sides $\therefore (x + y\,i)^2 = \frac{1 + 4\sqrt{3}\,i}{1 - 4\sqrt{3}\,i}$

$\therefore (x + y\,i)^2 = \frac{1 + 4\sqrt{3}\,i}{1 - 4\sqrt{3}\,i} \times \frac{1 + 4\sqrt{3}\,i}{1 + 4\sqrt{3}\,i} = \frac{\left(1 + 4\sqrt{3}\,i\right)^2}{1 + 48} = \frac{\left(1 + 4\sqrt{3}\,i\right)^2}{49} = \left(\frac{1 + 4\sqrt{3}\,i}{7}\right)^2$

$\therefore x + y\,i = \pm\frac{1 + 4\sqrt{3}\,i}{7} = \pm\left(\frac{1}{7} + \frac{4\sqrt{3}}{7}\,i\right)$ $\therefore x = \pm\frac{1}{7}$, $y = \pm\frac{4\sqrt{3}}{7}$

Example 6

If $(x + y\,i)^2 (1 + 2\,i) = 11 + 2\,i$, find the real values of x , y

Solution

$\because (x + y\,i)^2 (1 + 2\,i) = 11 + 2\,i$ $\therefore (x + y\,i)^2 = \frac{11 + 2\,i}{1 + 2\,i}$

$\therefore (x + y\,i)^2 = \frac{(11 + 2\,i)(1 - 2\,i)}{(1 + 2\,i) + (1 - 2\,i)} = \frac{15 - 20\,i}{5}$ $\therefore (x + y\,i)^2 = 3 - 4\,i$

$\therefore x^2 - y^2 + 2xy\,i = 3 - 4\,i$ $\therefore x^2 - y^2 = 3$ (1)

, $2xy = -4$ (2)

By squaring (1) , (2) and adding : $\therefore (x^2 + y^2)^2 = 25$ $\therefore x^2 + y^2 = 5$ (3)

By adding (1) , (3) : $2x^2 = 8$ $\therefore x^2 = 4$

$\therefore x = \pm 2$

By subtracting (1) from (3) : $\therefore 2y^2 = 2$ $\therefore y^2 = 1$

$\therefore y = \mp 1$

From (2) : $xy < 0$

$\therefore x$, y have different signs $\therefore x = 2$, $y = -1$ or $x = -2$, $y = 1$

Example 7

If $x = 8 - 6i$, find the values of $x^{-\frac{3}{2}}$

Solution

$\because x = 8 - 6i$, $x^{\frac{3}{2}} = x \times x^{\frac{1}{2}}$

Assume $x^{\frac{1}{2}} = a + bi$ by squaring both sides

$\therefore x = (a + bi)^2$ $\therefore 8 - 6i = a^2 - b^2 + 2abi$

$\therefore a^2 - b^2 = 8$ (1)

, $2ab = -6$ (2)

By squaring (1) , (2) and adding : $\therefore (a^2 + b^2)^2 = 100$

$\therefore a^2 + b^2 = 10$ (3)

By adding (1) , (3) : $\therefore 2a^2 = 18$ $\therefore a^2 = 9$ $\therefore a = \pm 3$

By subtracting (1) from (3) : $\therefore 2b^2 = 2$ $\therefore b^2 = 1$ $\therefore b = \mp 1$

From (2) : $ab < 0$ $\therefore$ a and b have different signs

$\therefore x^{\frac{1}{2}} = \pm (3 - i)$ $\therefore x^{\frac{3}{2}} = x \times x^{\frac{1}{2}} = \pm (8 - 6i)(3 - i)$

$\therefore x^{\frac{3}{2}} = \pm (18 - 26i)$ $\therefore x^{-\frac{3}{2}} = \pm \frac{1}{18 - 26i}$

$\therefore x^{-\frac{3}{2}} = \pm \frac{18 + 26i}{(18 - 26i)(18 + 26i)} = \pm \frac{18 + 26i}{1000} = \pm (0.018 + 0.026i)$

Example 8

If $x \in \mathbb{C}$, find the solution set of the equation $x^2 - (2 + i)x - (1 + 5i) = 0$

Solution

$\because x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ where $a = 1$, $b = -(2 + i)$, $c = -(1 + 5i)$

$\therefore x = \frac{(2 + i) \pm \sqrt{(2 + i)^2 + 4(1 + 5i)}}{2}$ $\therefore x = \frac{(2 + i) \pm \sqrt{3 + 4i + 4 + 20i}}{2}$

$\therefore x = \frac{(2+i) \pm \sqrt{7+24i}}{2}$

Let $\sqrt{7+24i} = \ell + m i$ $\quad\quad$ $\therefore 7 + 24i = \ell^2 - m^2 + 2\ell m i$

$\therefore \ell^2 - m^2 = 7$ (1)

and $2\ell m = 24$ (2)

By squaring (1) , (2) and adding : $\therefore (\ell^2 + m^2)^2 = 625$

$\therefore \ell^2 + m^2 = 25$ (3)

By adding (1) , (3) : $\therefore 2\ell^2 = 32$ $\quad\quad$ $\therefore \ell^2 = 16$ $\quad\quad$ $\therefore \ell = \pm 4$

By subtracting (1) from (3) : $\therefore 2m^2 = 18$ $\quad\quad$ $\therefore m^2 = 9$ $\quad\quad$ $\therefore m = \pm 3$

$\therefore \sqrt{7+24i} = \pm (4 + 3i)$ $\quad\quad$ $\therefore x = \frac{(2+i) \pm (4+3i)}{2}$

$\therefore x = \frac{6+4i}{2} = 3 + 2i$ or $x = \frac{-2-2i}{2} = -1 - i$

$\therefore$ The solution set $= \{3 + 2i , -1 - i\}$

Example 9

Find the exponential form (Euler form) of the number :

$z = \frac{(1+\sqrt{3}\,i)(i-\sqrt{3})^5}{8i^3}$ **, then find the cubic roots of the number z in the exponential form.**

Solution

$z = \frac{(1+\sqrt{3}\,i)(i-\sqrt{3})^5}{8i^3} \times \frac{i}{i} = \frac{(i-\sqrt{3})(i-\sqrt{3})^5}{8} = \frac{1}{8}(i-\sqrt{3})^6$

Put $i - \sqrt{3} = -\sqrt{3} + i$ in the trigonometric form

$\therefore r = \sqrt{x^2 + y^2} = \sqrt{3+1} = 2$

, $\because x = -\sqrt{3} < 0$, $y = 1 > 0$

$\therefore -\sqrt{3} + i$ lies in the 2nd quadrant

$\therefore \theta = 180° + \tan^{-1}\left(\frac{y}{x}\right) = 180° + \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right) = 180° + (-30°) = 150°$

$\therefore i - \sqrt{3} = 2(\cos 150° + i \sin 150°) = 2\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)$

$\therefore (i-\sqrt{3})^6 = 2^6\left(\cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6}\right)^6 = 64(\cos 5\pi + i \sin 5\pi)$

$= 64(\cos \pi + i \sin \pi)$

$\therefore z = 8(\cos \pi + i \sin \pi) = 8 e^{\pi i}$ (The exponential form)

$\therefore z^{\frac{1}{3}} = 8^{\frac{1}{3}} e^{\frac{(\pi + 2\pi n) i}{3}}$ where $n = 0 , 1 , -1$

at $n = 0$, $\therefore z_1 = 2 e^{\frac{\pi}{3} i}$

at $n = 1$, $\therefore z_2 = 2 e^{\pi i}$

at $n = -1$, $\therefore z_3 = 2 e^{\frac{-\pi}{3} i}$

Example 10

Use De Moivre's theorem to express each of $\cos 2\theta$, $\sin 2\theta$ in terms of $\sin\theta$, $\cos\theta$

Solution

$\because \cos 2\theta + i \sin 2\theta = (\cos\theta + i \sin\theta)^2 = \cos^2\theta - \sin^2\theta + 2 \sin\theta \cos\theta \, i$

and from the equality of two complex numbers

$\cos 2\theta = \cos^2\theta - \sin^2\theta$, $\sin 2\theta = 2 \sin\theta \cos\theta$